Gecko Inspired Adhesives

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About Gecko and its ability

Geckos, found **worldwide** in warm regions, inhabit diverse terrains like forests, deserts, mountainous areas, grasslands, and urban spaces. The **most common** species, **Tokay Gecko** , thrives in tropical and rocky habitats. Their adaptability to various environments makes them **vital for pest control** and ecological balance.

Geckos can **stick to surfaces** using specialized toe pads covered in microscopic hairs called **setae**. These setae create **van der Waals forces**, allowing geckos to adhere to smooth, vertical, or **even inverted** surfaces without sticky substances. This remarkable adhesion is non-permanent and **doesn't leave residue** when they detach.

Limitations of conventional adhesive technologies

Glue leaves **permanent residue** and can damage surfaces

Adhesive tapes **lose adhesion after repeated use**

Suction cups require **smooth, airtight surfaces**

Velcro systems are **bulky and noisy**

These limitations motivated researchers to explore biological adhesion mechanisms found in nature.

Learning from Nature

Biomimicry is the practice of **designing technologies** inspired by **natural biological systems** that have evolved efficient solutions over **millions of years.**

- Engineers study biological systems to solve **complex engineering challenges**
- One remarkable example is the gecko's ability to adhere to **smooth vertical surfaces** and ceilings
- This natural mechanism has **inspired the development** of gecko-inspired adhesive technologies

The Gecko’s Extraordinary Ability

Geckos possess a remarkable ability to climb **smooth vertical surfaces** and even move **upside down on ceilings** with ease.

- They can adhere to surfaces such as **glass, metal, and plastic** without the use of glue, suction, or liquids.
- The adhesive force generated by their feet is **strong enough to support their entire body weight**, yet it can be **released instantly while walking**.
- This ability is made possible by **millions of microscopic contact structures on their feet**, which allow extremely close interaction with surfaces.

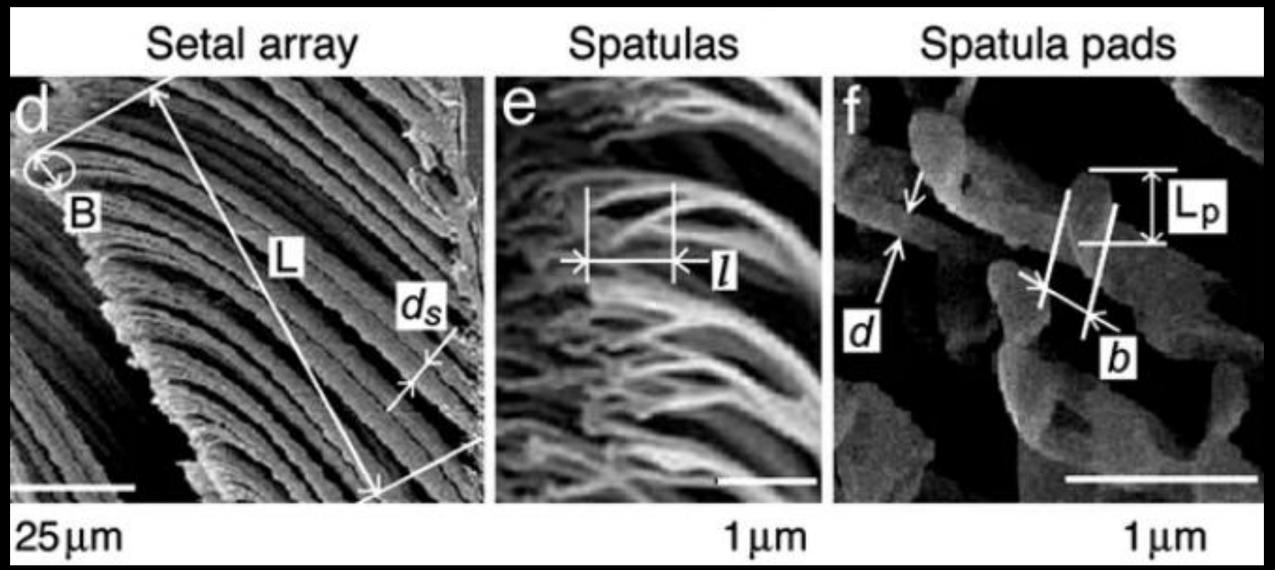
How does it do it ?

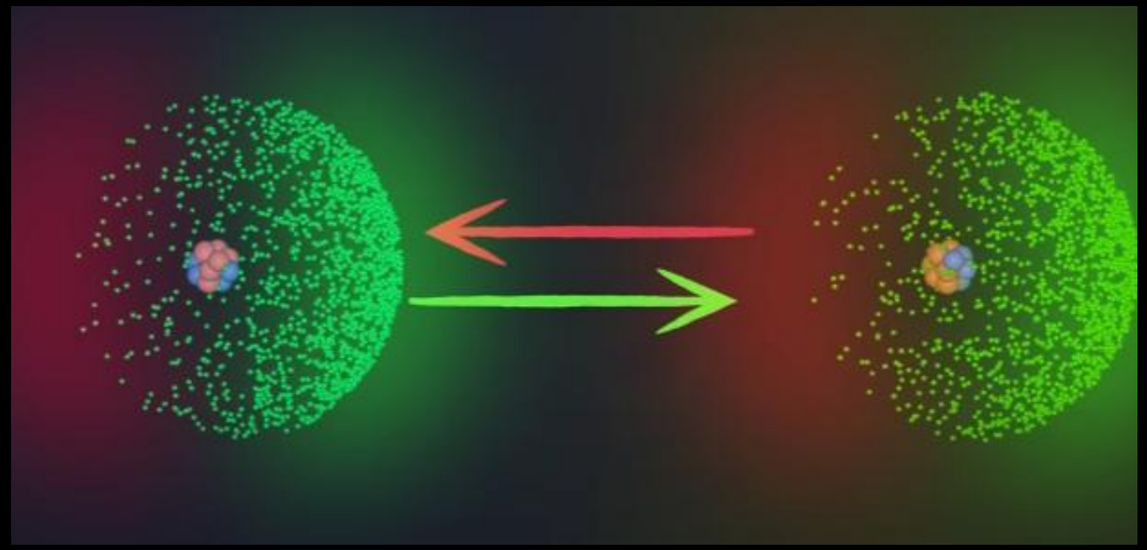
The gecko sticks with **temporary** attractive forces between molecules called **Van der Waals** forces. They are easily formed and broken again, and only occur over **short distances**, like when the molecules on the gecko's feet are very close to the **molecules of the wall**. The distance between the molecules has to be very tiny. The molecules are approximately **0.3 – 0.6 nanometers** apart.

https://www.acs.org/education/celebrating-chemistry-editions/2020-ncw/gecko

The **spatulae** on the bottom of geckos' feet interact with the surface through van der Waals forces. Van der Waal forces are **intermolecular attractions** that create temporary dipoles. Dipoles are caused by unequal areas of electron density around molecules, which, if **covalently bonded**, can either be **polar or nonpolar**, depending on the shape of the molecule.

The force results from a **transient shift in electron density**. Specifically, the electron density may temporarily shift to be greater on one side of the nucleus. This shift **generates a transient charge** which a nearby atom can be attracted to or repelled by. The force is **repulsive at very short distances**, reaches **zero** at an **equilibrium distance** characteristic for each atom, or molecule, and becomes **attractive** for distances **larger** than the **equilibrium distance.**



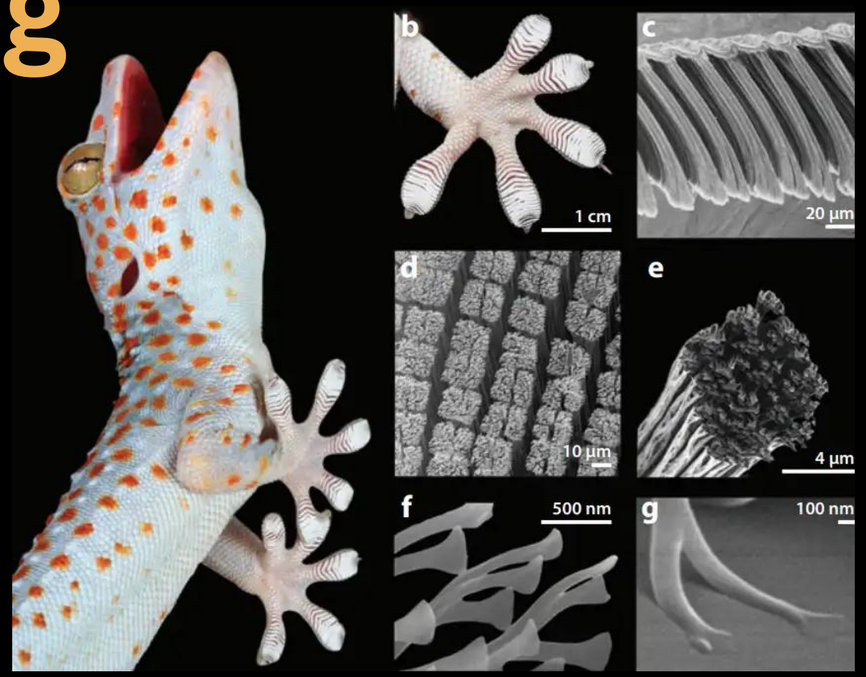


https://www.acs.org/content/dam/acsorg/education/students/highschool/chemistryclubs/infographics/geckos-infographic.pdf

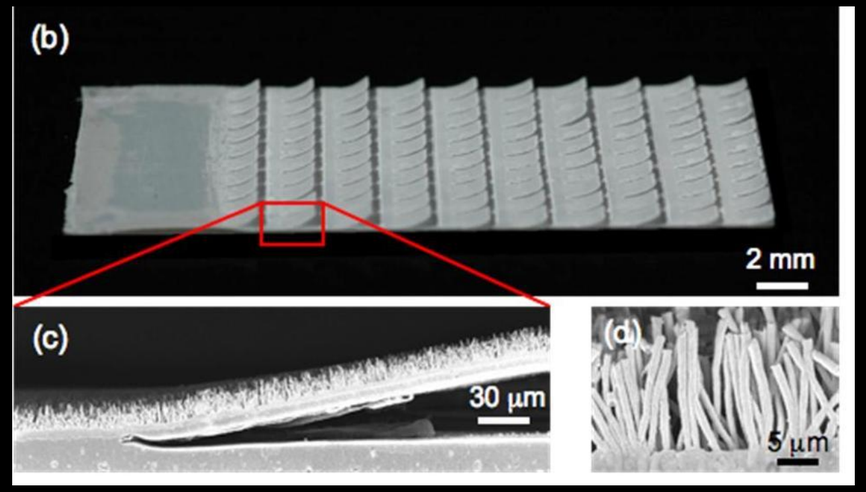
https://en.wikipedia.org/wiki/Van_der_Waals_force

From Biology to engineering

Scientists studied the **microscopic structure of gecko feet** to understand how geckos generate strong yet reversible adhesion.



- Gecko toe pads contain **millions of microscopic hairs called setae** that increase surface contact.
- These structures allow the foot to come **extremely close to surfaces**, enabling intermolecular interactions.
- Engineers replicate these structures using synthetic **microfibers and nanomaterials**.



This research led to the development of gecko-inspired adhesive technologies.

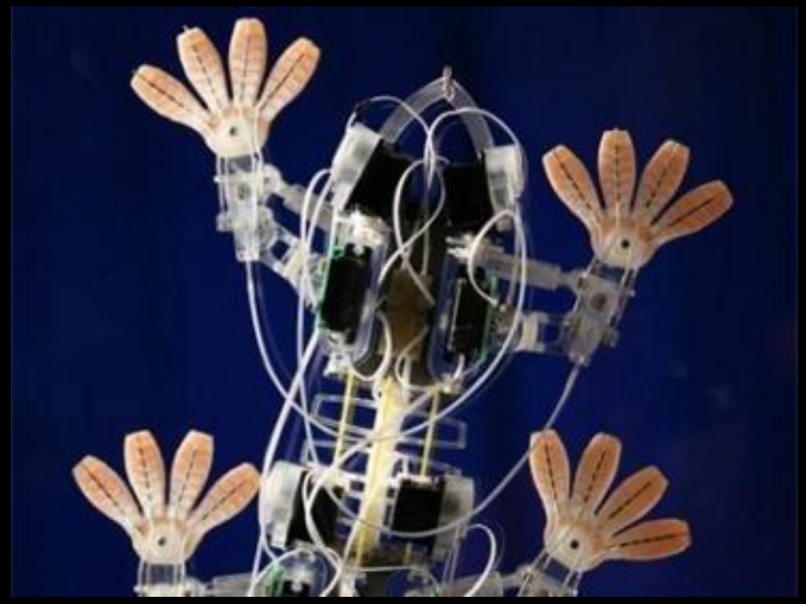
The **most prominent** application

Gecko tape is a next-gen adhesive inspired by how geckos climb walls. Instead of glue, it uses **tiny hair-like structures** to create a strong, reusable grip—without leaving any residue.

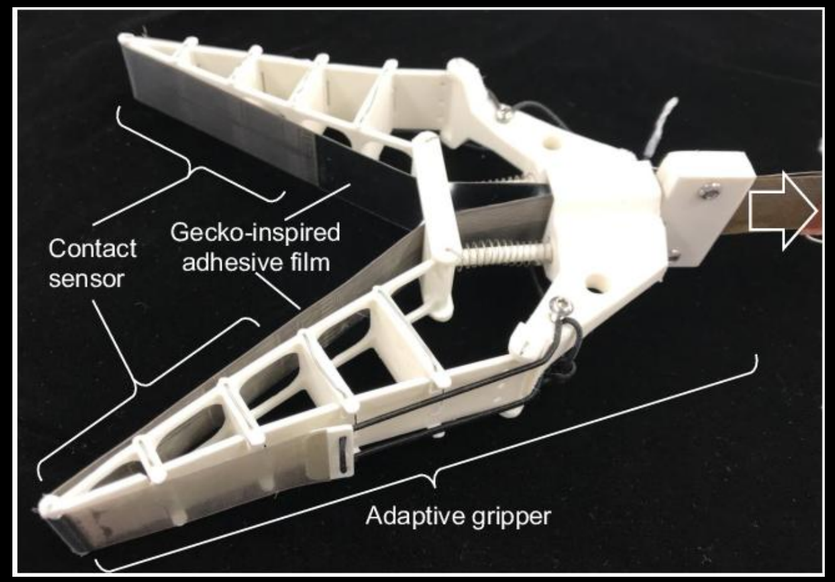
This high-tech tape works on various surfaces, even in **space and underwater**, making it a game-changer for **robotics, aerospace, and medicine.**

Building on the Gecko Tape

Climbing Robots



Precision Grippers



Where did it all start?

In **2002**, Kellar Autumn’s experiments revealed **geckos' ability to stick to surfaces** was due to microscopic hairs called **setae on their toes**. Using force measurement tools, the team discovered these hairs generate van der Waals forces, enabling **adhesion without sticky substances.**This breakthrough revolutionized biomimicry and inspired **advanced adhesive technologies.**

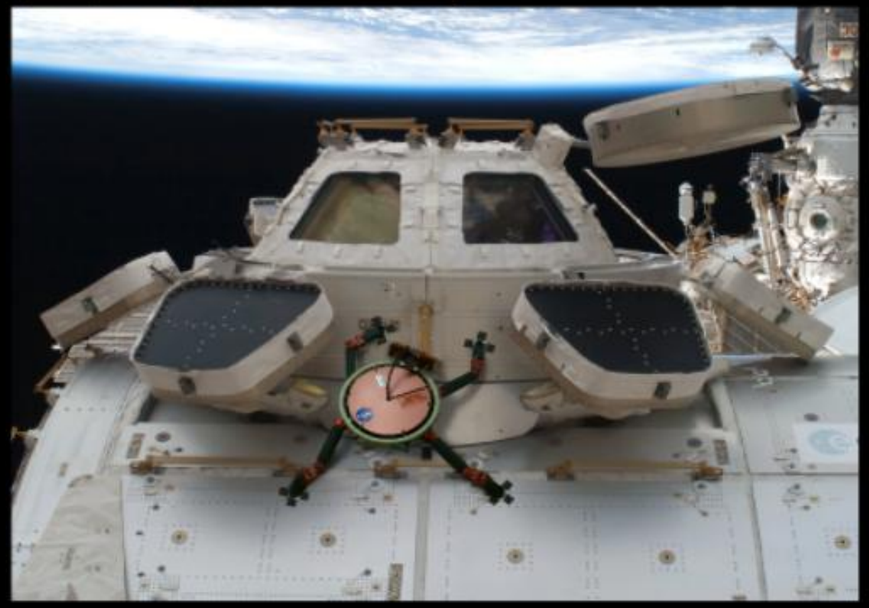
Meet stickybot

The gecko-inspired adhesive technology was **first implemented around 2004**. Researchers at Stanford University, led by **Mark Cutkosky**, developed a gecko-inspired **climbing robot** named Stickybot, which could scale smooth surfaces **using synthetic setae**. This marked the **first practical application** of gecko adhesion principles in robotics and biomimetic engineering.

https://news.mit.edu/2009/stickybot-092509

Building on the Stickybot

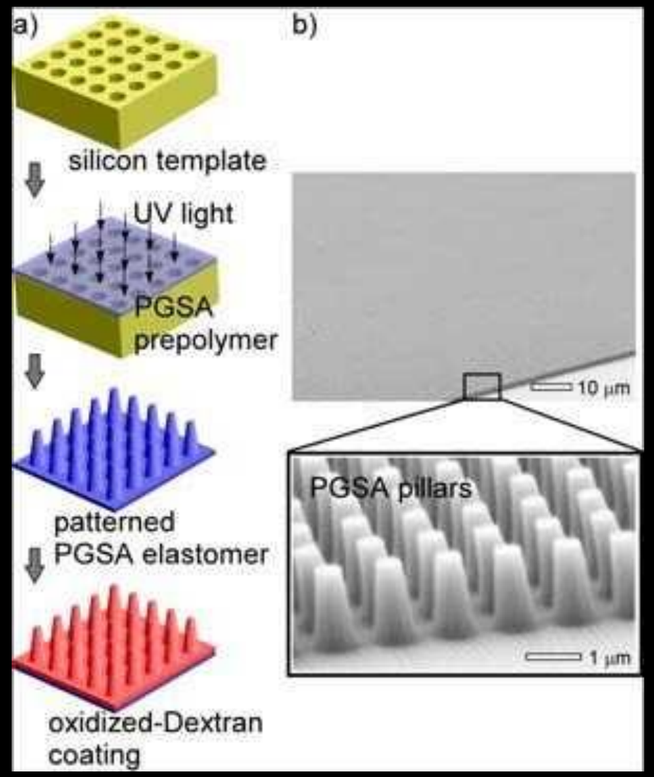
Stickybot's innovative **adhesion technology** has led to significant **advancements in robotics** and other fields. Its design inspired the development of gecko-inspired **grippers**, used for precise handling and manipulation of objects. **Climbing robots** which can scale vertical surfaces, aiding in inspection, maintenance, and search-and-rescue operations. In the **medical field**, it led to bandages that adhere securely to skin without using **traditional adhesives**. The principles of Stickybot have also been used in the tools for **space exploration** to attach in environments where **traditional adhesives fail**, such as zero gravity.



Innovations In The Medical Field

Gecko-inspired adhesion technology has led to **non-invasive wound dressings** that adhere securely to the skin without irritation. This technology is also used in surgical robots for precise **tissue handling** and biodegradable adhesive strips for dermal tissue applications, offering a safer alternative to sutures. These innovations improve patient comfort and reduce healing time.

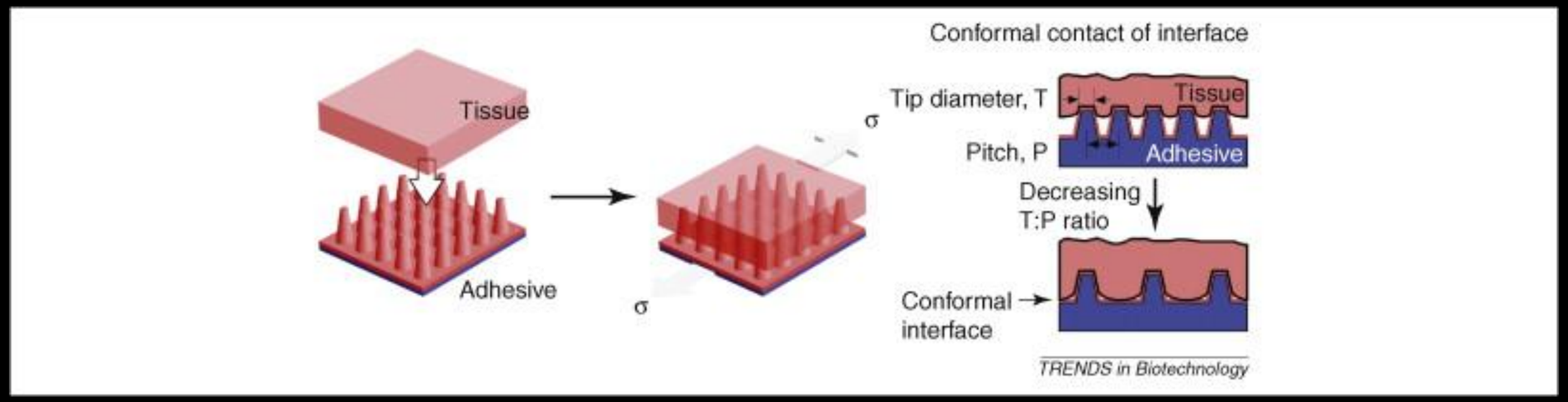
Gecko-inspired medical adhesives were first **implemented in 2008** by researchers at Harvard University, particularly for bandages and surgical tools. These adhesives have been used in various applications to reduce pain and enhance recovery.



https://www.nature.com/search?q=Gecko+Inspired+Bandages&order=relevance

Advantages of the Bandage

- **Non-Irritating Adhesion**: Adheres without causing skin irritation, unlike regular adhesives.
- **Gentle Removal**: Can be removed easily without pain or damaging healthy tissue.
- **Breathability**: Allows the skin to breathe, reducing infection risks and promoting faster healing.
- **Better Wound Healing**: Provides a stable environment, aiding quicker and more effective healing.
- **Biodegradability**: Made from eco-friendly, biodegradable materials.
- **Flexibility and Comfort**: Conforms to body contours, providing greater comfort during movement.



Disadvantages of the bandage

- **Cost**: The advanced materials and technology involved in creating gecko-inspired adhesives may result in higher production costs, making them more expensive than traditional bandages.
- **Limited Availability**: These specialized bandages may not be as widely available as conventional ones, limiting access in certain regions.
- **Durability**: While effective for many uses, the adhesion strength may decrease over time or under harsh conditions, potentially affecting their reliability.
- **Complex Manufacturing**: Producing the synthetic setae for the adhesive is complex and may require specialized equipment or processes, which could limit scalability.
- **Sensitivity to Surfaces**: Gecko-inspired adhesives work best on clean, smooth surfaces, and may not perform as well on dirty or rough skin, limiting their effectiveness in certain situations.
- **Allergies or Sensitivities**: Though less likely, some individuals may have an allergic reaction to the materials used in the synthetic setae, causing irritation or discomfort.

Astrobee: NASA's Gecko-Inspired Robot

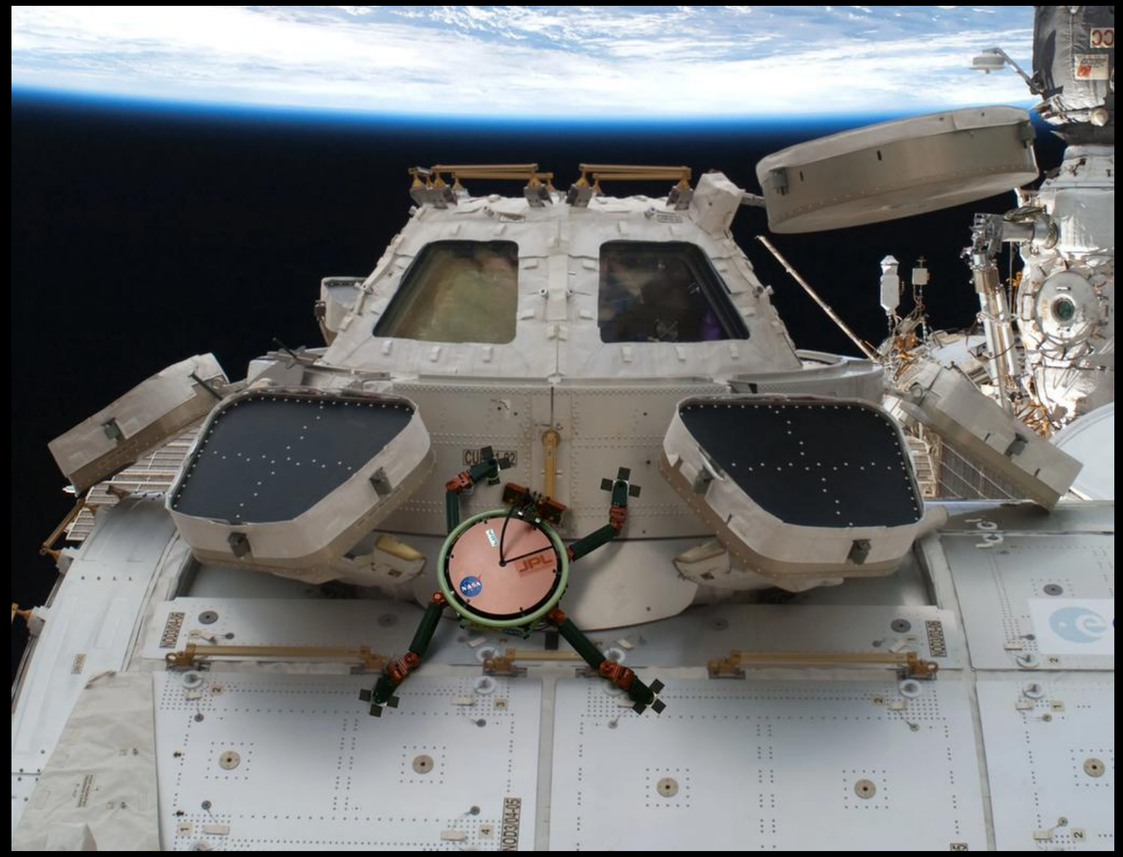
NASA's **Astrobee** is a free-flying robotic system used aboard the **International Space Station** (ISS). Inspired by gecko adhesion, Astrobee incorporates gecko-inspired adhesives for **precise handling** and **attachment** to surfaces without the need for **mechanical clamps** or traditional adhesives. This technology enhances Astrobee’s ability to perform **delicate tasks** such as assisting astronauts and conducting experiments. It also draws from the gecko's ability to cling to and manipulate objects safely in microgravity.

https://www.nasa.gov/astrobee/

NASA's Gecko Grippers

NASA's Gecko Grippers are inspired by the adhesive abilities of geckos. They use **microstructures** on their surface to mimic the gecko's setae, enabling them to stick to and release surfaces using van der Waals forces. This technology allows for secure object handling in **microgravity environments**, making it ideal for **satellite servicing** and **space station maintenance**.

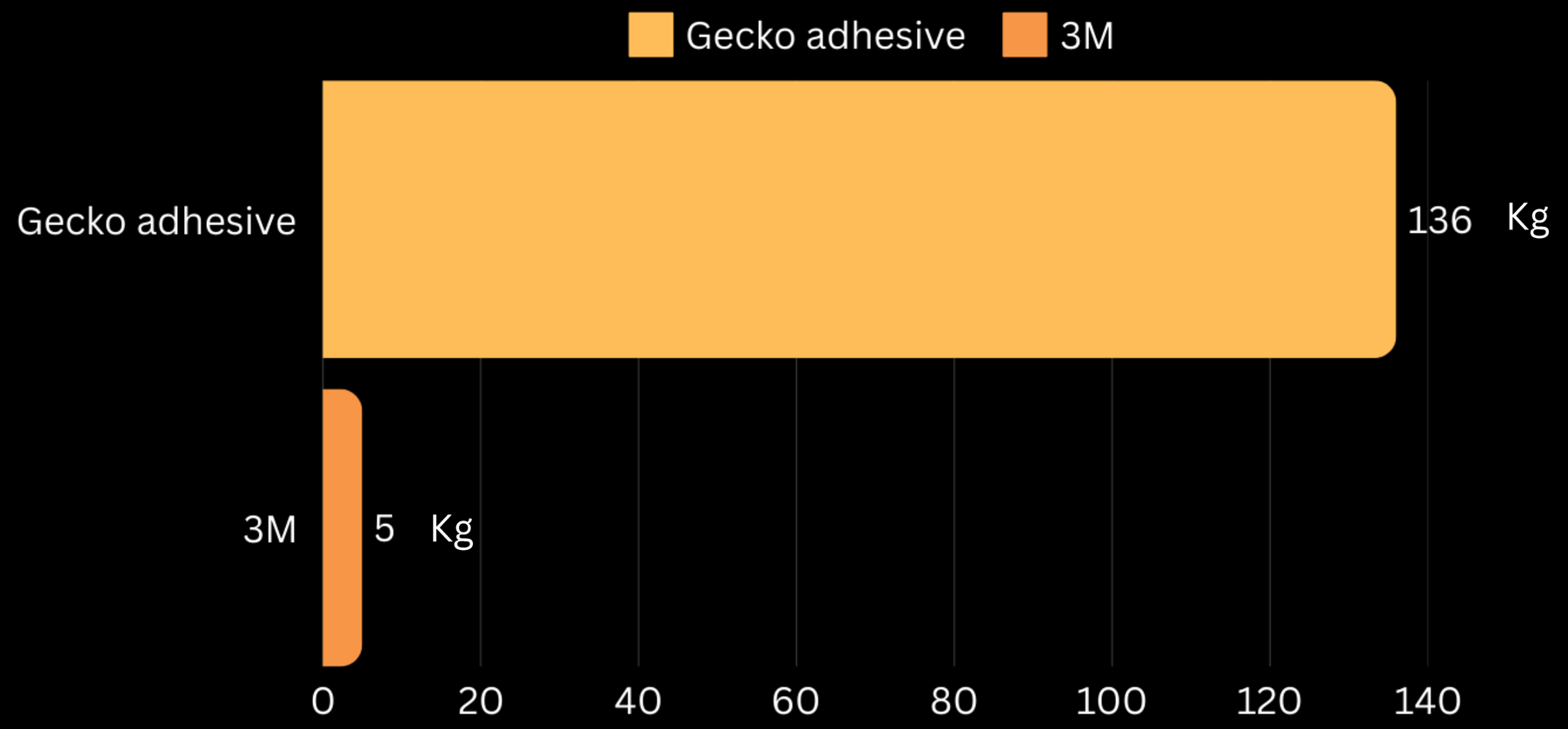
This natural mechanism inspired NASA to create a synthetic adhesive for robotic grippers that could operate in microgravity, where traditional gripping methods fail. The goal was to develop a **reliable**, **reusable** gripping technology for space exploration tasks, such as handling satellites and assisting astronauts in **zero-gravity environments.**



https://www.nasa.gov/missions/station/gecko-grippers-moving-on-up/

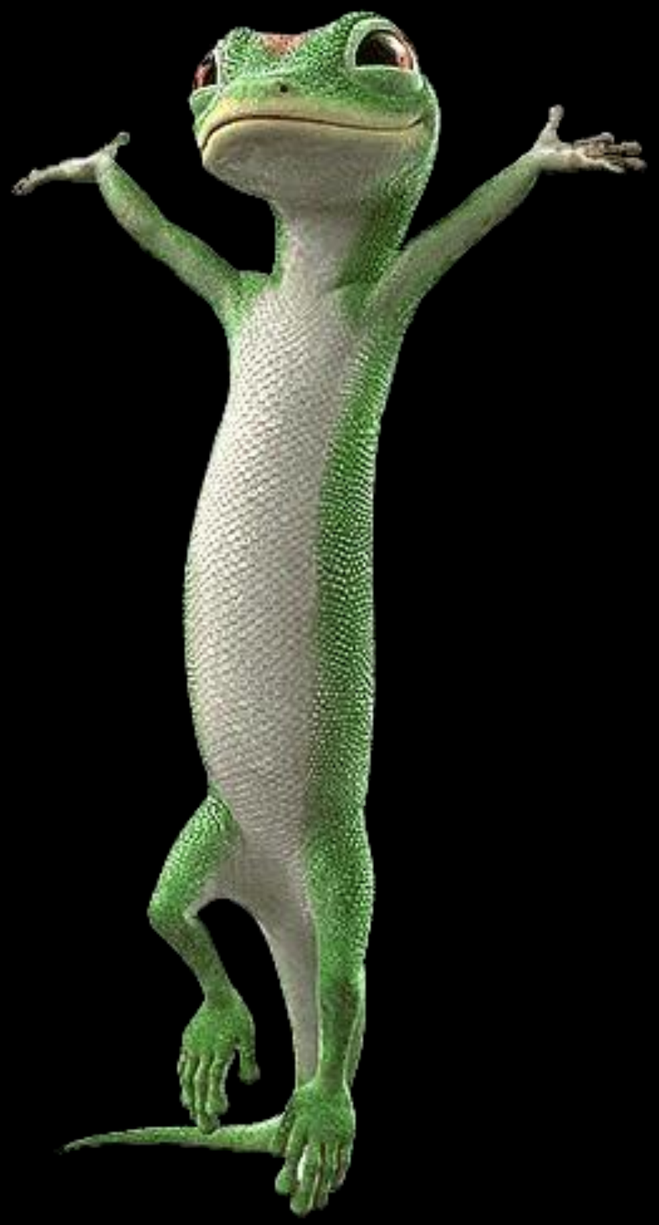
COROLLA

Weight carrying capacity



If its so great,

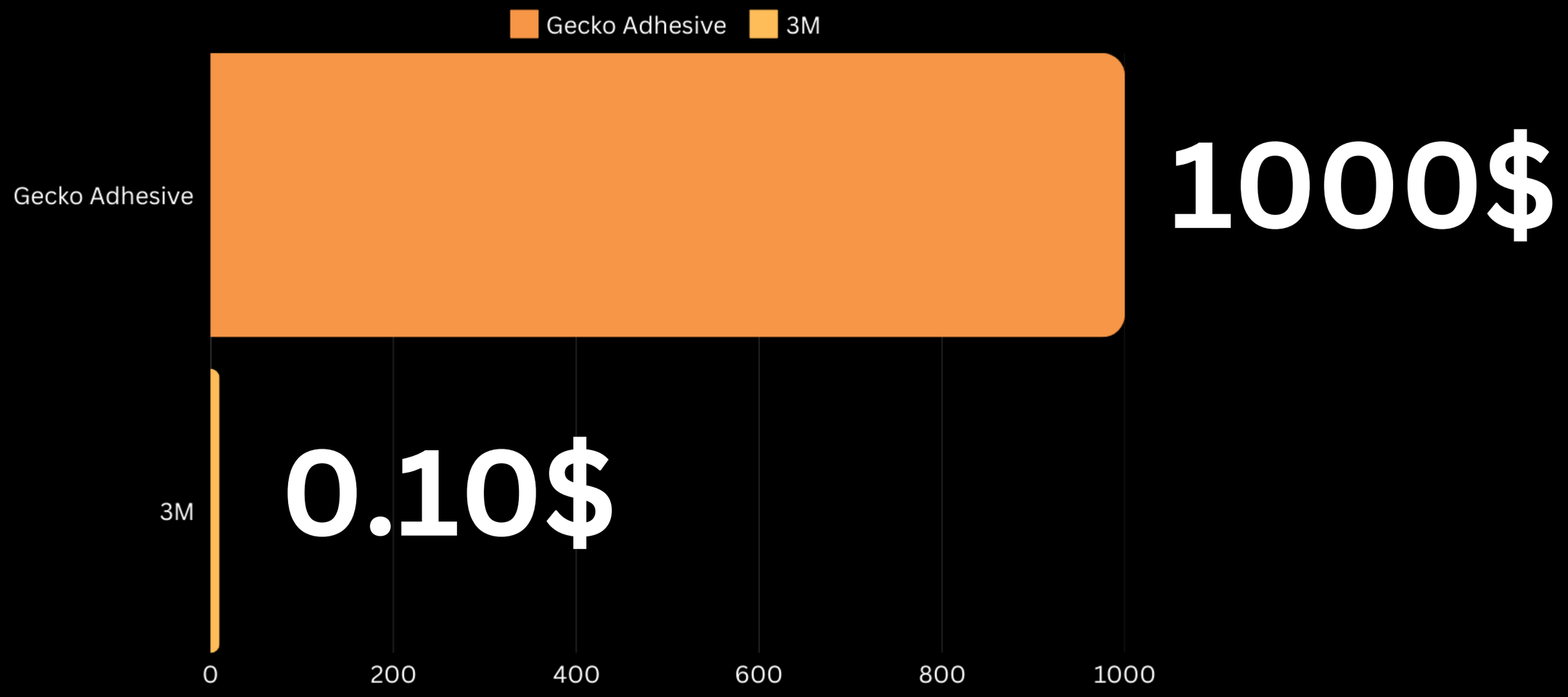
Then why not??



Well....

MONEY!

Cost Comparison



Limitations of the Gecko Gripper

1. **Surface Dependency:** The grippers work best on smooth, clean surfaces. Dust, debris, or irregular textures can reduce their effectiveness.
2. **Force Limitations:** Gecko-inspired adhesion relies on van der Waals forces, which can only generate a limited gripping strength, making it unsuitable for very heavy loads.
3. **Temperature Sensitivity:** Extreme temperatures in space may affect the performance and longevity of the synthetic adhesive.
4. **Complex Manufacturing:** Creating the microstructures that mimic gecko setae is technically challenging and costly.
5. **Wear and Tear:** Repeated use can degrade the microstructures over time, reducing adhesive performance.

2002

Autumn and colleagues publish findings in The **Journal of Experimental Biology**, further elucidating the mechanism behind **gecko adhesion.**

2005

Mark Cutkosky's team at Stanford University develops **"Stickybot,"** a robot utilizing synthetic setae to mimic gecko-like **climbing abilities.**

2008

Researchers from **Harvard University** introduce gecko-inspired medical adhesives designed for bandages and surgical applications, aiming to **reduce pain and enhance recovery.**

2012

The company Felsuma unveils "**Geckskin**," an adhesive technology replicating the gecko's tendon system, **offering strong adhesion** without residue.

2019

NASA deploys the **Astrobee** robots aboard the International Space Station (ISS), to assist with object manipulation in **microgravity** environments.

2021

NASA tested **gecko-inspired robotic** grippers aboard the ISS.

Future Potential

1. Nanotechnology

Advancements in nanotechnology will enable the creation of even stronger and more durable gecko-inspired adhesive tapes. Researchers are exploring the **use of nanomaterials**, such as carbon nanotubes, to enhance the adhesive properties of the tapes. These nanomaterials can provide **greater surface area** for contact and stronger van der Waals forces, resulting in improved adhesion.

2. Bio-Inspired Design

The study of gecko feet continues to inspire new adhesive tape designs. Researchers are learning more about the intricate structure of gecko setae and spatulae, which will result in the development of more sophisticated and adaptable adhesive tapes. This bio-inspired approach will lead to **more efficient** and **versatile** adhesive solutions.

And to finally...

wrap it all up

Conclusion

1. Geckos demonstrate a remarkable **natural adhesion mechanism**

2. Scientists discovered this ability comes from **millions of microscopic contacts generating van der Waals forces.**

3. Engineers have replicated these principles to develop **advanced adhesive technologies for robotics, medicine, and space exploration.**